# ANALYSIS OF COVARIANCE

TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 22

5-MINUTE REVISION

Source: Chapter 22 of the textbook.

## 1. QUICK REVISION · 1

- ANCOVA combines treatment comparison with regression adjustment.
- A covariate should be relevant to the response and should not be caused by the treatment being compared.
- The common-slope model assumes the same regression slope across treatments; examine treatment-by-covariate interaction.
- Independent errors, appropriate linearity and common residual variance are required for the stated model; normal errors support the usual exact tests.
- Adjustment compares treatment means at a common covariate value.

## 2. QUICK REVISION · 2

- Estimating one covariate slope uses one additional error degree of freedom.
- A useful covariate can improve precision, but improvement is not guaranteed.
- Adjustment does not turn an observational comparison into a randomized experiment.
- Use pair-specific standard errors when covariate means differ.

## 3. IMPORTANT FORMULAE · 1

**For the CRD with one covariate and a common within-treatment slope:**

$$Y_{ij} = \mu + \tau_i + \beta(X_{ij} - \bar{X}) + e_{ij}, \qquad \hat{\beta} = \frac{E_{xy}}{E_{xx}}$$

**Here $E_{xx}$, $E_{xy}$ and $E_{yy}$ are the within-treatment corrected sums of squares and products defined in the chapter, with $E_{xx} > 0$.**

$$E = E_{yy} - \frac{E_{xy}^2}{E_{xx}}, \qquad MSE_{adj} = \frac{E}{n - v - 1}$$

**With $n$ observed responses, $v$ treatments and one fitted covariate slope, the CRD error degrees of freedom are $n - v - 1$. For the complete RBD with $r$ blocks, use the within-block-and-treatment quantities and error degrees of freedom $(r - 1)(v - 1) - 1$.**

$$\bar{Y}_{i,adj} = \bar{Y}_i - \hat{\beta}(\bar{X}_i - \bar{X})$$

$$F_{Treatment} = \frac{(E_1 - E)/(v - 1)}{MSE_{adj}}$$

## 4. IMPORTANT FORMULAE · 2

Here $E_1$ is the residual sum of squares from the reduced model without treatment effects, retaining the covariate and, in RBD, the blocks. $E$ is from the full model. The treatment comparison is therefore adjusted for the covariate.

## 5. SELF-CHECK FROM THE CHAPTER

1. ANCOVA combines analysis of variance with _____. **[ Regression ]**
2. A measured explanatory variable used for statistical adjustment is called a _____. **[ Covariate ]**
3. The common-slope ANCOVA model assumes equal regression _____ across treatments. **[ Slopes ]**
4. A covariate measured before treatment application is called a _____ covariate. **[ Pre-treatment ]**